Architecture and stack

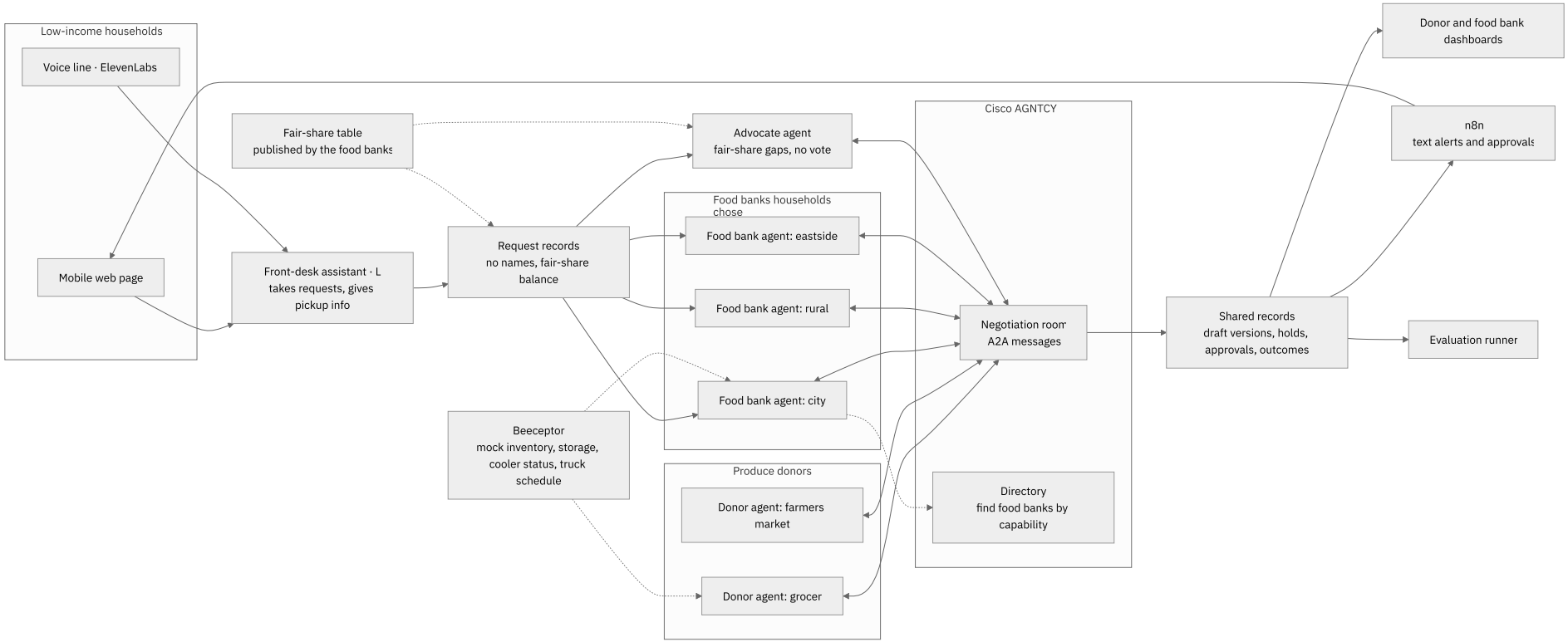
What runs where, and what each sponsor does.

> **AFTER THIS LESSON**
> You can sketch the architecture from memory and give each sponsor's role in one line.

The pieces

- **Households:** mobile web page or voice line, then the front-desk assistant, which saves request records with no names.
- **Agents:** one for each donor and each food bank, plus the advocate. They negotiate in the AGNTCY room over A2A messages.
- **Directory:** finds food banks by capability, for example when a cooler fails.
- **Shared records:** draft versions, holds, approvals, and outcomes.
- **n8n:** turns records into text alerts and approval requests.
- **Dashboards and the evaluation runner:** read the records but can't change any decision.
- **Beeceptor:** stands in for each organization's own systems: inventory, storage, cooler status, and truck schedule.

> Diagram on the next page (landscape): HarvestNet architecture. From docs/MVP_DESIGN.md.


Low-income households
Voice line · ElevenLabs
Mobile web page
Front-desk assistant · L
takes requests, gives
pickup info
Fair-share table
published by the food banks
Request records
no names, fair-share
balance
Advocate agent
fair-share gaps, no vote
Food banks households
chose
Food bank agent: eastside
Food bank agent: rural
Food bank agent: city
Beeceptor
mock inventory, storage,
cooler status, truck
schedule
Produce donors
Donor agent: farmers
market
Donor agent: grocer
Cisco AGNTCY
Negotiation room
A2A messages
Directory
find food banks by
capability
Shared records
draft versions, holds,
approvals, outcomes
Donor and food bank
dashboards
n8n
text alerts and approvals
Evaluation runner

Sponsors, one line each

- **Cisco AGNTCY:** A2A messages and Directory discovery. Skip SLIM group chat, which is marked experimental.
- **ElevenLabs:** the voice line, with English and Spanish read-back.
- **n8n:** alerts, approvals, and the "can't make it" release.
- **Beeceptor:** mock systems, plus the switch that makes a cooler fail.
- **AWS:** hosting. **Claude API:** the agents and the assistant.

None of them decides anything (ADR 0004).

See the AGNTCY app-sdk README and the A2A specification.

On AWS

One EC2 machine running Docker Compose. Each organization's agent gets its own container, alongside the web app and API, the AGNTCY Directory, and Postgres. There's one public HTTPS address, so n8n and ElevenLabs can reach the system, and keys are kept in Secrets Manager. n8n, Beeceptor, and ElevenLabs stay on the sponsors' own servers.

Models

Opus 5.5 for food banks and the advocate, Sonnet 5.5 for donors and the front desk, and no AI for scoring. This assignment is proposed.

Source: docs/MVP_DESIGN.md, sections 7, 13, 14, and 15.

Read next: docs/MVP_DESIGN.md, section 7 and its architecture diagram.

Previous: Lesson 10: Proving it works · **Next:** Lesson 12: Leading the build

Quick lookup: HarvestNet at a glance.

Anything unclear? Ask me. I'm your teacher for this, and I can re-explain any part in different words.

Source of truth: docs/MVP_DESIGN.md, CONTEXT.md, and docs/adr/. If this lesson and those disagree, they win.